Backend plan from now to hack day

Shane and Atif · Thursday 5 November 2026 · replaces the task order in the playbook and in docs/backend-plan.md. Hack days are Wednesday 11 to Friday 13 November.

Where we are

- **Built and on GitHub:** the interface, the FastAPI server, the engine, the offline reasoner, 25 backend tests, 3 chat tests and the interface gate.
- **Merged today:** Atif's fixes for citations, ranking and location checks. Shane's real-data check, the fix for Gemma stopping mid-answer, the setup guide and AGENTS.md.
- **Not done yet:** nothing has run on the real HPC data, and Gemma has not been tested. Five bugs are open, and each one is a task below.
- **Targets:** the first real-data answer on Friday 6 Nov. Version 1 with Gemma on real data by Saturday 7 Nov. Feature freeze on Tuesday 10 Nov in the evening.

How we work: three gates

Every task follows the same path. Shane runs each check with Claude Code. Nothing reaches main without both gates.

Step	Who	What happens
1. Plan	Task owner	Write five lines or fewer: task ID, files, approach, the test you will add, estimate.
Gate 1	Shane + Claude	Shane pastes the plan to Claude. Claude checks it against AGENTS.md and the code. Go, or change it.
2. Build	Task owner	On a branch named owner/task-id-name. Add a test that fails before the fix and passes after.
3. Laptop tests	Task owner	Run laptop tests 1 to 5 (section 5). Paste the pass counts into the pull request.
Gate 2	Shane + Claude	Shane asks Claude to review the pull request. Fix what it finds. Then Shane merges.
4. HPC	Shane	git pull on the HPC, restart, run the HPC tests (section 6).
Gate 3	Both	Any change to API fields, chat events or the store layout is agreed before anyone writes it.

Message templates to send Shane:

```
Gate 1 - A1 - plan
Files: backend/agent.py
Approach: <two lines>
Test: <name and what it checks>
Estimate: 2 h

Gate 2 - A1 - PR #<number>
Tests: unittest 26 pass, node 3 pass, check_ui passed, eval 10/10, browser checklist done
```

Schedule

Estimates are working hours. Waiting on Ruthan or for an HPC slot is not counted.

Day	Shane	Atif	Checkpoint by the end of the day
Thu 5 Nov evening	S0 HPC setup (30 min) S1 real-data check (0.5 to 2 h)	A0 pull, tests (20 min) A2 winding fix (30 min) Gate 1 plan for A1	Check output reviewed with Claude
Fri 6 Nov	S2 first real run (1 h) S3 ask Ruthan	A1 opportunity (2 h) A3 safe rebuild (1.5 h)	First real-data answer on the HPC
Sat 7 Nov	S3 rerun with Ruthan's inputs (1 h) A4 with Atif on the HPC	A4 Gemma on the HPC (2 to 3 h) Gate 1 plan for A5	Version 1: Gemma answers on real data
Sun 8 Nov	A5 chat side (30 min) S4 question bank (start)	A5 follow-ups (3 to 4 h) Gate 1 plan for A6	Follow-up questions work
Mon 9 Nov	S4 Gemma eval, Ruthan scores (3 h)	A6 blind well test (4 to 6 h)	Blind test ready to show
Tue 10 Nov	S5 freeze, backup video, rehearsal (3 h)	Bug fixes from the eval only	Freeze at 8 pm
Wed 11 to Fri 13 Nov	Mentor feedback, rehearsals	Mentor feedback fixes	Present on Fri 13

Total	Hours
Atif	about 14 to 19 h (A0 to A6)
Shane	about 10 to 12 h (S0 to S5, plus the chat side of A5 and the HPC half of A4)

Order of priority if time runs short

- **Must have for the demo:** S0 to S3, A1, A3 and A4. These give a real-data answer that a geologist will accept.
- **Should have:** A5 follow-ups and S4 eval. Judges will ask a follow-up during Q&A.
- **Nice to have:** A6 blind well test. Show it only if the hidden outcome is provably excluded. Otherwise leave it out.

Tasks: Atif

A0 Pull main and align Codex

Atif · 20 min

Do	git checkout main, git pull. Read AGENTS.md, which Codex also reads on its own. Run laptop tests 1 to 5.
Done when	25 Python tests and 3 chat tests pass, and the app loads with MOCK=1.

A1 "Opportunity" means an undrilled lead

Atif · 2 h

Problem	With a drilled lead selected, such as L1, "What is the geological opportunity, and why?" calls L1 the opportunity. An opportunity has not been drilled. The answer without a selection is already correct.
Do	When the question asks for the opportunity or where to drill next and the selected lead is drilled, answer about the best undrilled lead. Say the selected lead is a drilled calibration lead. Questions that name a lead, such as "Tell me about L1", keep using that lead.
Files	backend/agent.py (rule planner and offline writer). The Gemma planner prompt in backend/prompts.py if it makes the same mistake.
Test to add	A drilled lead is selected and the headline question is asked. The answer names the undrilled lead and does not call the drilled lead the opportunity. "Tell me about L1" still uses L1.

A2 "Winding" only when the channel actually winds

Atif · 30 min

Problem	backend/agent.py, around line 281, always says "elongate and winding". The engine measured a sinuosity of 1.0 for a lead on the synthetic survey, which means a straight body.
Do	Say "winding" only when sinuosity is at least the engine's channel threshold (engine/leads.py Params.sinuosity_channel, 1.3). Otherwise say "elongate", or "compact" if elongation is low too.
Test to add	A lead with sinuosity 1.0 never gets "winding". A lead with 1.5 does.

A3 Rebuild the store without deleting the working one

Atif · 1.5 h

Problem	engine/precompute.py deletes the output folder before it builds. A crash or a bad window during the demo leaves the app with no data.
Do	Build into a sibling folder such as copilot_store.new. On success, swap it into place and remove the old store. On failure, leave the old store untouched and print why.
Files	engine/precompute.py. Tell Shane first, because engine/ is his area.
Test to add	A build that raises halfway leaves the old store intact with no .new folder. A good build replaces the store and leaves no .old folder.

A4 Gemma on the HPC

Atif with Shane · 2 to 3 h

Do	Book a slot with Shane on the remote desktop. With LLM=gemma on the real store, run the 10 questions of eval/questions_template.csv (Shane fills in real well names on the HPC) and the browser checklist. Count answers whose footer says "Gemma couldn't plan" or "stopped partway". Time each answer. Fix the prompts in backend/prompts.py, and the timeout in config.py (LLM_TIMEOUT_S) if needed.
Done when	At least 8 of 10 eval answers are written by Gemma without a planner fallback. Each answer finishes in under 30 s, and no answer reports numbers that aren't in the evidence.
Note	Eval results from real data stay on the HPC. Write down only the counts and timings, never the answers.

A5 Follow-up questions

Atif, chat side by Shane · 3 to 4 h

Problem	"Why?" or "What about the seal?" after an answer about L3 is treated as a new question with no context.
Gate 3 proposal	/api/chat accepts an optional history field: the last two turns as {question, lead}. The chat sends it, and the server never stores it, so sessions stay separate. Agree on this before building.
Do	If the question names no lead and reads as a follow-up (short, "why", "what about", "and the"), reuse the previous lead and topic. "The seal" maps to the risk tools for that lead. A new lead selection resets the context.
Tests to add	"Why?" after an L3 answer fetches L3 evidence. "What about the seal?" fetches L3 risks. After L1 is selected, "Why?" uses L1.

A6 Blind well test, for real

Atif · 4 to 6 h

Problem	hide_wells is accepted but ignored. The interface hides the well on screen only, so the AI can still see the result.
Do	Build a per-request view of the store that removes the hidden well's result, its log summary and its result-based labels from every tool: the wells list, well summary, lead evidence, calibration averages, similar leads, the planner's well list and the cache key. Seismic values at the well stay available. Then set hide_wells to true in /api/health. Shane removes the "UI mask only" label.
Test to add	With a well hidden, its result text and log numbers appear in no tool output and no prompt sent to Gemma. Changing only its result changes nothing the agent sees before the reveal.

Tasks: Shane

S0 HPC setup

Shane · 30 min

Do	Follow docs/team-setup.md, HPC part 1: clone into /data/epic20/APGCE/apgce_team_06, environment in team storage, pip install.
Done when	python scripts/check_real_data.py starts without import errors.

S1 Check the real data

Shane · 0.5 to 2 h

Do	Run python scripts/check_real_data.py. Screenshot the output to Claude. Fix any FAIL together: header bytes, well coordinates or missing results. The script reads headers and a sample of traces only.
Done when	All 9 volumes are ok, the wells land inside the survey, and there is a provisional window.

S2 First real run and Gemma smoke test

Shane · 1 h

Do	Run precompute with the provisional window, then render_instrument, then the server. Ask three questions: the headline question, "Rank the leads" and one about a well. Screenshot each answer to Claude.
Watch	The precompute reads all 9 volumes into memory. If it is killed or very slow, tell Claude, and it will switch it to read one volume at a time.
Done when	/api/health shows mock false, llm_mode gemma, llm_reachable true. Answers show "Gemma 4" in their footer.

S3 Ruthan's inputs

Shane with Ruthan · 1 h plus waiting

Do	Get the reservoir window in ms, read on an inline in DUG Insight. If the well list has no results, copy data/well_results.example.csv to data/well_results.csv on the HPC and have Ruthan fill it in. That file stays on the HPC. Rerun precompute.
Done when	Ruthan agrees that the wells and leads on the map match DUG Insight, and that the scorecard reasons make sense.

S4 Question bank and scoring

Shane with Ruthan · 3 h

Do	On the HPC, write 30 questions in eval/questions.csv, using the columns of eval/questions_template.csv. The file holds Ruthan's answers about real data, so git ignores it and it stays on the HPC. Include the three judge-style questions, rankings, wells, look-alikes, risks, follow-ups and two questions with no evidence. Run the eval with Gemma on the HPC. Ruthan scores the sheet.
Done when	An average score is recorded, and the three lowest answers are turned into tasks.

S5 Freeze and backup

Shane · 3 h

Do	Tuesday 10 Nov, 8 pm: no new features. Run MOCK=1 python frontend/build_standalone.py for the offline demo file. Record a backup screen video of the full demo on the HPC. Two rehearsals.

Tests on both laptops

Run all of these before every pull request. On Windows PowerShell, set MOCK once per terminal with $env:MOCK="1" instead of the MOCK=1 prefix.

#	Command	Must show	Who
1	python -m unittest discover -s tests -p "test_*.py"	All pass (25 today, more as tasks add tests)	Both
2	node --test tests/test_chat.mjs	3 pass	Both
3	python scripts/check_ui.py	All checks passed (contrast, colours, spacing, no slop, no CDN)	Both
4	MOCK=1 python -m engine.precompute MOCK=1 uvicorn backend.app:app --port 8800	App loads at localhost:8800, then the browser checklist below	Both
5	MOCK=1 python eval/run_eval.py --questions eval/questions_mock.csv --tag <branch>	10 answers, none empty, no tracebacks	Both
6	Fake SEG-Y check (docs/team-setup.md, laptop step 5)	All checks passed, and the server answers	Whoever touched engine/
7	MOCK=1 python frontend/build_standalone.py	The file is written and opens with no network	Shane, after frontend changes

Browser checklist, about 3 minutes

- The welcome screen loads and opens the workspace.
- Clicking a lead updates the map, the scorecard and the section below the map.
- "What is the geological opportunity, and why?" streams an answer with evidence tags.
- Clicking an evidence tag opens its dialog and pins the point on the map. Escape closes it.
- "Rank the leads", with a lead selected, shows a table of all leads.
- "Is there gas at inline 5000, crossline 2050?" answers "Location unavailable".
- Executive mode, the blind well test, the committee memo and the light theme all open without errors.
- The browser console shows no red errors.

Tests on the HPC (Shane, after every merge)

```
cd /data/epic20/APGCE/apgce_team_06/exploration-copilot-team6
source /data/epic20/APGCE/apgce_team_06/copilot-env/bin/activate
export COPILOT_STORE_DIR=/data/epic20/APGCE/apgce_team_06/copilot_store
git pull
```

When	Command	Must show
engine/ changed	python scripts/check_real_data.py	No FAIL lines
engine/ changed	python -m engine.precompute --t0 <ms> --t1 <ms> python -m engine.render_instrument	Store written, leads and wells counted
Always	uvicorn backend.app:app --host 127.0.0.1 --port 8800	Server starts
Always	curl -s http://127.0.0.1:8800/api/health	"mock":false, "llm_mode":"gemma", "llm_reachable":true
Always	Browser checklist in the HPC's Firefox	Footers say "Gemma 4". Wells and leads match DUG Insight.
backend/ changed	python eval/run_eval.py --questions eval/questions.csv --tag gemma-<date>	Once S4 exists. The sheet stays on the HPC.

Never

- Commit seismic, well or store files, screenshots of real maps, HPC logins, the HPC manual, the RustDesk key or tokens. The repo is public.
- Copy challenge data, or answers built from it, off the HPC.
- Push to main or merge your own pull request. Codex and Claude work on branches only.
- Invent numbers. Every number in an answer or on screen comes from the store or from the user's own inputs.
- Edit code on the HPC without pushing it straight away, or run a second server there.
- Start a task without a gate 1 go, or change an API field or chat event without gate 3.